Data Sheet

PAW3327DB-TWQU: Optical Gaming Navigation Chip

General Description

PAW3327DB-TWQU is PixArt Imaging's latest state of the art entry gaming mouse chip. It is packaged in an 8-pin staggered dual in-line package (DIP). It provides great gaming experience with the features of high speed of 220ips and resolution of 6,200 cpi to fulfill today's gamers' need. PAW3327DB-TWQU chip along with LM31-LNG lens or LM33-LSG, LED clip, and the recommended HSDL-4261 IR LED forms a complete and compact mouse tracking system to achieve optimum performance. It is suitable for wired gaming navigation system.

Key Features

- Small form factor PDIP 8L molded lead-frame package
- Operating Voltage: 3.00V - 3.60V
- High speed motion detection 220ips (typical) and acceleration 30g (max).
- Selectable resolutions up to 6200cpi
- Four-wire serial port interface
- Internal oscillator — no clock input needed
- Angle snapping
- On-chip LED driver
- Minimal number of passive components

Applications

- Wired Gaming Optical Mouse
- Trackball application

Key Parameters

Parameter	Value
Power Supply Voltage (V)	VDD: 3.00V – 3.60V
Interface	4-wire Serial Peripheral Interface
Supply Current @ VDD = 3.3V *Note: includes LED current*	Run: 14 mA Power Down: 98 uA
Resolution (cpi)	Up to 6200
Tracking Speed (ips)	220
Acceleration (g)	30
Package Type (mm)	8L PDIP, 9.9 x 12.85 x 6.1

Ordering Information

Part Number	Package Type
PAW3327DB-TWQU	8L PDIP
LM31-LNG	Trim Lens
LM33-LSG	Small Trim Lens

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1.0 Introduction

1.1 Overview

PAW3327DB-TWQU is based on Optical Navigation Technology, which measures changes in position by optically acquiring sequential surface images (frames) and mathematically determining the direction and magnitude of movement. PAW3327DB-TWQU contains an Image Acquisition System (IAS), a Digital Signal Processor (DSP), and a four-wire serial port. The IAS acquires microscopic surface images via the lens and illumination system. These images are processed by the DSP to determine the direction and distance of motion. The DSP calculates the ΔX and ΔY relative displacement values. An external microcontroller reads and translates the ΔX and ΔY information from the chip serial port into PS2, USB, or RF signals before sending them to the host PC.

Note: Throughout this document PAW3327DB-TWQU is referred to as the chip.

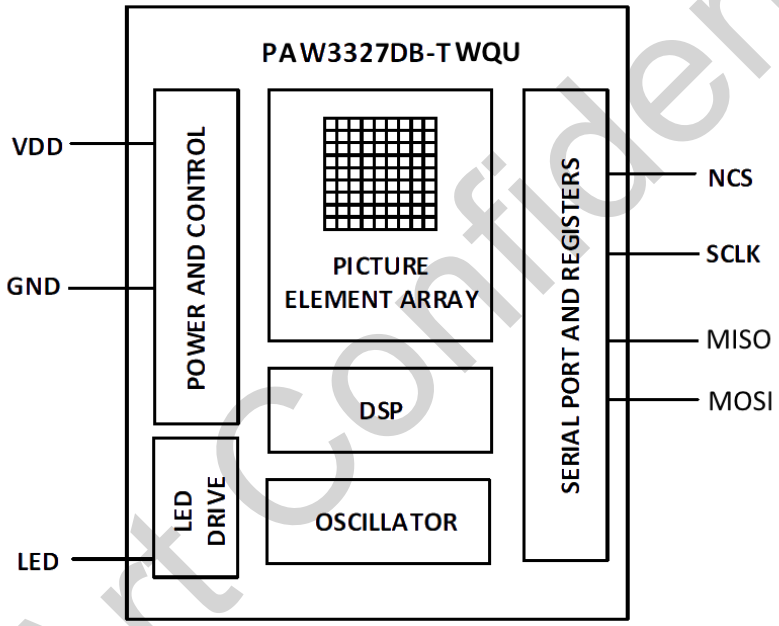


Figure 1. Functional Block Diagram

1.2 Terminology

Term	Description
DSP	Digital Signal Processing
LED	Light Emitting Diode
NCS	Chip Select
VDD	Supply voltage
SCLK	Serial Clock
MISO	Serial Data Out
MOSI	Serial Data In
GND	Ground

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Pins Description

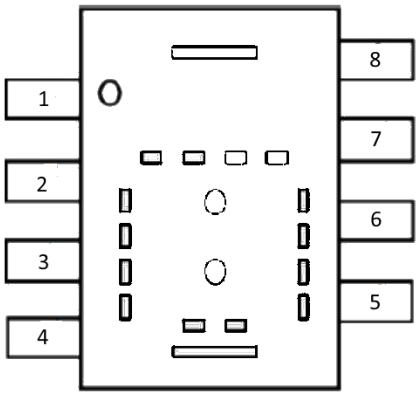


Figure 2. Pin Configuration

Table 1. PAW3327DB-TWQU Pins Description

Pin No.	Name	Type	Description
1	MISO	Ouput	Serial data output
2	XY_LED	Input	LED Illumination Control input
3	MOTION	Output	Motion interrupt
4	NCS	Input	Chip select (active low)
5	SCLK	Input	Serial data clock
6	GND	Gnd	Ground
7	VDD	Power	Input power supply
8	MOSI	Input	Serial data input

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2.0 Operating Specifications

2.1 Absolute Maximum Ratings

Stresses greater than those listed under "Absolute Maximum Ratings" may cause permanent damage to the device. These are the stress ratings only and functional operation of the device at these or any other condition beyond those indicated for extended period of time may affect device reliability.

Table 2. Absolute Maximum Ratings

Parameters	Symbol	Min.	Max.	Unit	Notes
Storage Temperature	T_S	-40	85	°C	
Lead Solder Temperature	T_{SOLDER}		260	°C	For 10 seconds, 1.6mm below seating plane
Supply Voltage	V_{DD}	-0.5	3.6	V	Including V_{NA} of 100 mV_{pp}
ESD	ESD_{HBM}		2	kV	All pins, Human Body Model
Input Voltage	V_{IN}	-0.5	3.6	V	All digital pins
Output Latch up Current	mA		100		All digital pins

Notes:

1. Stresses greater than those listed under "Absolute Maximum Ratings" may cause permanent damage to the device. These are the stress ratings only and functional operation of the device at these or any other condition beyond those indicated for extended period of time may affect device reliability.
2. The maximum ratings do not reflect eye-safe operation.
3. The inherent design of this component causes it to be sensitive to electrostatic discharge. The ESD threshold is listed above. To prevent ESD induced damage, take adequate ESD precautions when handling this product

2.2 Recommended Operating Conditions

Table 3. Recommended Operating Conditions

Description	Symbol	Min.	Typ.	Max.	Unit	Notes
Operating Temperature	T_A	0		40	°C	
Power Supply Voltage	V_{DD}	3.00	3.30	3.60	V	Including supply noise
Power Supply Rise Time	t_{RT}	0.15		20	ms	0 to VDD
Supply Noise (Sinusoidal)	V_{NA}			100	mV_{p-p}	10kHz – 50MHz
Serial Port Clock Frequency	f_{SCLK}			2	MHz	Active drive, 50% duty cycle
Distance from Lens Reference Plane to Tracking Surface	Z	2.20	2.40	2.60	mm	
Speed	S		220		ips	
Acceleration	A			30	g	In run mode
Load Capacitance	C_{out}			100	pF	MISO and MOTION pin

Note: PixArt does not guarantee the chip performance if the operating temperature is beyond the specified limit.

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2.3 Thermal Specifications

Table 4. Thermal Specifications

Parameters	Symbol	Min.	Typ.	Max.	Unit	Notes
Storage Temperature	T_S	-25	-	80	°C	
Lead-free Solder Temperature	T_P	-	-	260	°C	For 10 seconds, 1.6mm below seating plane for wave soldering

2.4 DC Characteristics

Table 5. DC Electrical Specifications

Parameters	Symbol	Min.	Typ.	Max.	Unit	Conditions
DC Supply Current in various modes Note: Includes I_{LED}	I_{DD_RUN}		14		mA	Average current consumption. MCU poll the motion registers at the 1ms interval.
Power Down Current	I_{DD_PD}		98		uA	
Input Low Voltage	V_{IL}			0.3* VDD	V	SCLK, MOSI, NCS
Input High Voltage	V_{IH}	0.7* VDD			V	SCLK, MOSI, NCS
Input Hysteresis	V_{I_HYS}		100		mV	SCLK, MOSI, NCS
Input High Leakage Current	I_{IH}		± 1	± 10	uA	Vin= 0.7*VDD, SCLK, MOSI, NCS
Input Low Leakage Current	I_{IL}		± 1	± 10	uA	Vin= 0.3*VDD, SCLK, MOSI, NCS
Output Low Voltage	V_{OL}			0.45	V	I_{OUT} = 1mA, MISO, MOTION
Output High Voltage	V_{OH}	VDD-0.45			V	I_{OUT} = -1mA, MISO, MOTION
Input Capacitance	C_{in}		50		pF	SCLK, MOSI, NCS

Note: All the parameters are tested under recommended operating conditions. Typical values at 25 °C, V_{DD} = 3.3 V & LED current = 15 mA

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2.5 AC Characteristics

Table 6. AC Electrical Specifications

Parameters	Symbol	Min.	Typ.	Max.	Unit	Conditions
Motion Delay After Reset	$t_{MOT-RST}$	50			ms	From reset to valid motion, assuming motion is present
Shutdown	t_{STDWN}			50	ms	From Shutdown mode active to low current
Wake From Shutdown	t_{WAKEUP}	50			ms	From Shutdown mode inactive to valid motion. Notes: A RESET must be asserted after a shutdown. Refer to section "Notes on Shutdown", also note $t_{MOT-RST}$
MISO Rise Time	t_{r-MISO}		50		ns	C_L = 100pF
MISO Fall Time	t_{f-MISO}		50		ns	C_L = 100pF
MISO Hold Time	$t_{hold-MISO}$	200			ns	Data held until next falling SCLK edge
MISO Delay After SCLK	$t_{DLY-MISO}$			120	ns	From SCLK falling edge to MISO data valid, no load conditions
MOSI Hold Time	$t_{hold-MOSI}$	200			ns	Amount of time data is valid after SCLK rising edge
MOSI Setup Time	$t_{setup-MOSI}$	120			ns	From data valid to SCLK rising edge
SPI Time Between Write Commands	t_{SWW}	30			µs	From rising SCLK for last bit of the first data byte, to rising SCLK for last bit of the second data byte.
SPI Time Between Write And Read Commands	t_{SWR}	20			µs	From rising SCLK for last bit of the first data byte, to rising SCLK for last bit of the second address byte.
SPI Time Between Read And Subsequent Commands	t_{SRW} tSRR	250			ns	From rising SCLK for last bit of the first data byte, to falling SCLK for the first bit of the address byte of the next command.
SPI Read Address-Data Delay	t_{SRAD}	4			µs	From rising SCLK for last bit of the address byte, to falling SCLK for first bit of data being read.
NCS Inactive After Motion Burst	t_{BEXIT}	500			ns	Minimum NCS inactive time after motion burst before next SPI usage
NCS To SCLK Active	$t_{NCS-SCLK}$	120			ns	From last NCS falling edge to first SCLK rising edge
SCLK To NCS Inactive (For Read Operation)	$t_{SCLK-NCS}$	120			ns	From last SCLK rising edge to NCS rising edge, for valid MISO data transfer

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Parameters	Symbol	Min.	Typ.	Max.	Unit	Conditions
SCLK To NCS Inactive (For Write Operation)	$t_{SCLK-NCS}$	20			µs	From last SCLK rising edge to NCS rising edge, for valid MOSI data transfer
NCS To MISO High-Z	$t_{NCS-MISO}$			250	ns	From NCS rising edge to MISO high-Z state
MOTION Rise Time	$t_{r-MOTION}$		50		ns	C_L = 100pF
MOTION Fall Time	$t_{f-MOTION}$		50		ns	C_L = 100pF
Input Capacitance	C_{in}		50		pF	SCLK, MOSI, NCS
Load Capacitance	C_L			100	pF	MISO, MOTION
Transient Supply Current	I_{DDT}			60	mA	Max supply current during the supply ramp from 0V to V_{DD} with min 150 us and max 20ms rise time. (Does not include charging currents for bypass capacitors)

Note: All the parameters are tested under recommended operating conditions. Typical values at 25 °C & VDD=3.3V

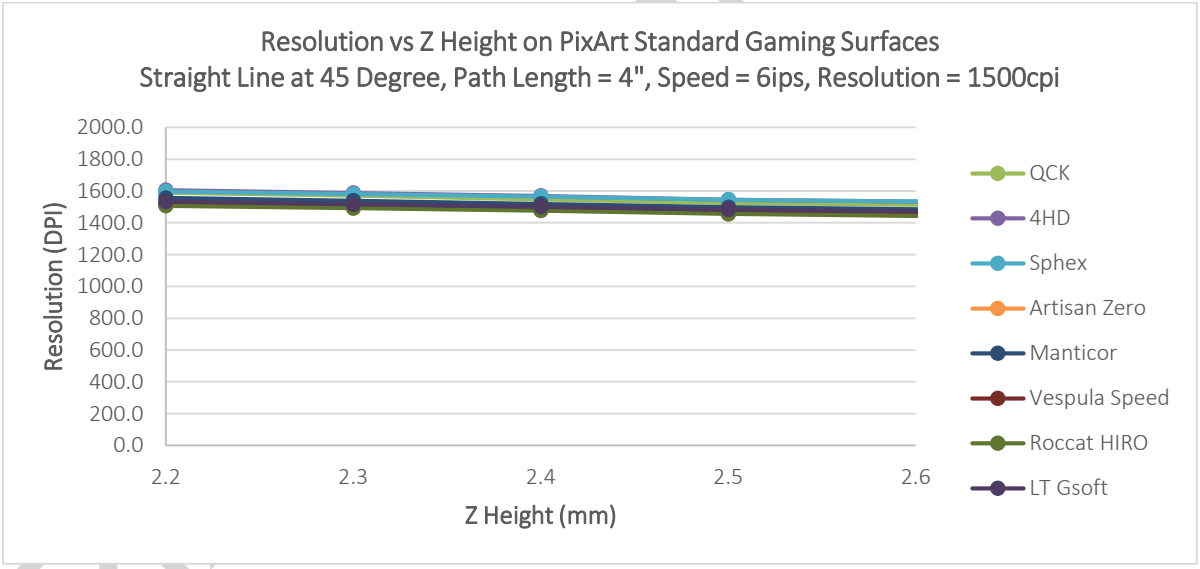


Figure 3. Mean Resolution vs. Z Height at resolution of 1500cpi

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3.0 Mechanical Specifications

3.1 Mechanical Dimension

Table 7. Package Dimensions

Parameters	Nominal	Min.	Max.	Unit
Package Body Dimension X	9.90	9.80	10.00	mm
Package Body Dimension Y	9.10	9.00	9.20	mm
Package Width (inclusive pins)	12.85	12.35	13.35	mm
Lead Length	5.15	5.05	5.25	mm
Lead Pitch	2.00	1.85	2.15	mm
Total Lead Count	8	-	-	-
Lead Offset	1.00	-	-	mm
Lead Width	0.50	0.40	0.60	mm
Hole Diameter	0.70	0.65	0.75	mm
Center of Hole from edge of body X	4.55	4.45	4.65	mm
Center of Hole from edge of body Y	3.92	3.82	4.02	mm

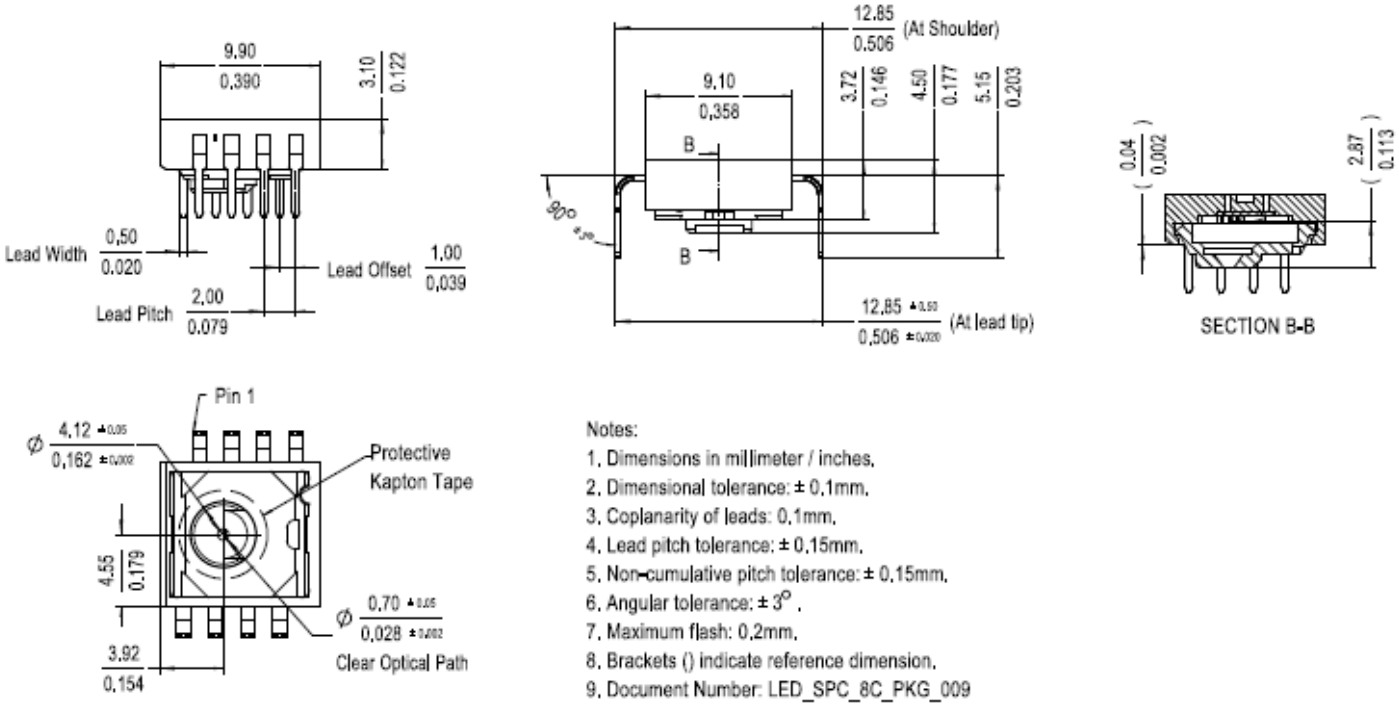


Figure 4. Package Drawing Outline

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3.2 Assembly Drawings

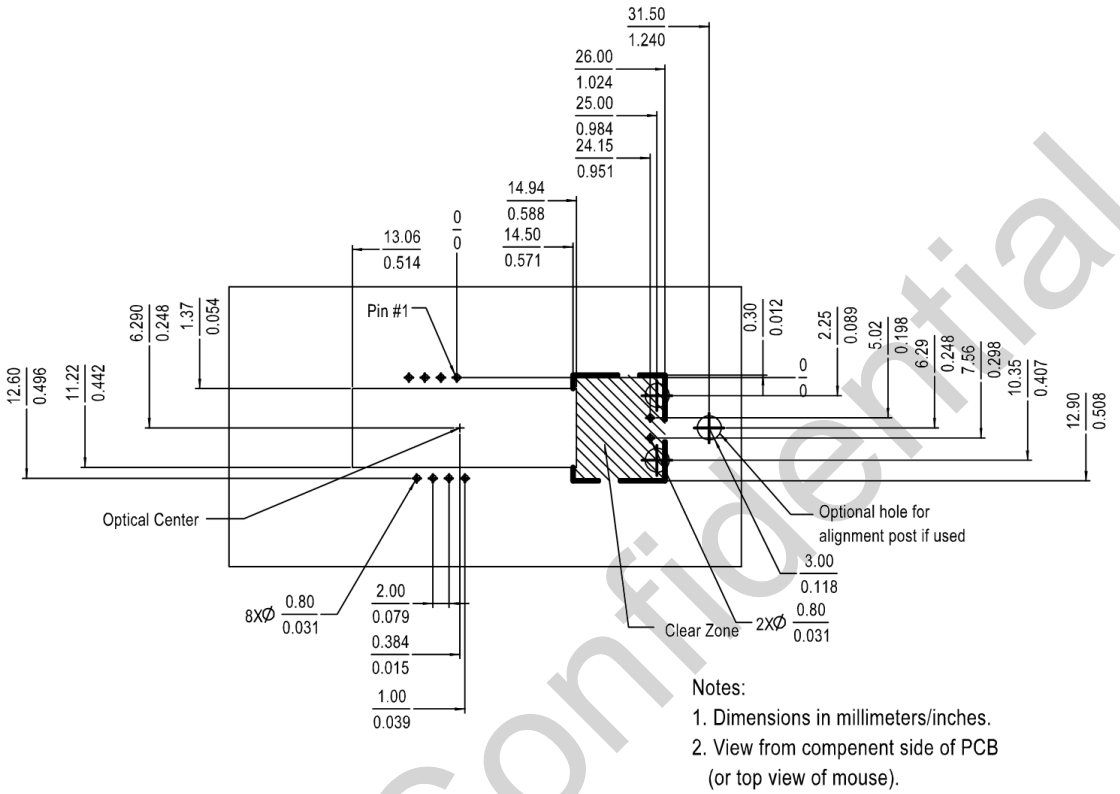


Figure 5. Recommended chip orientation, mechanical cutouts & spacing (Top View)

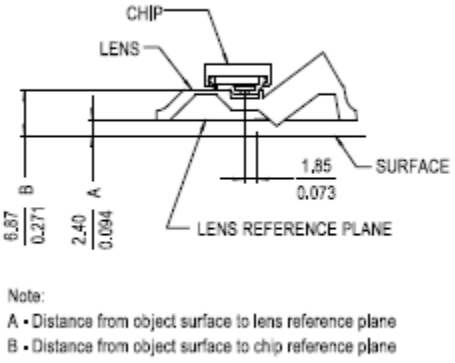


Figure 6. Distance from Lens Reference Plane to Surface

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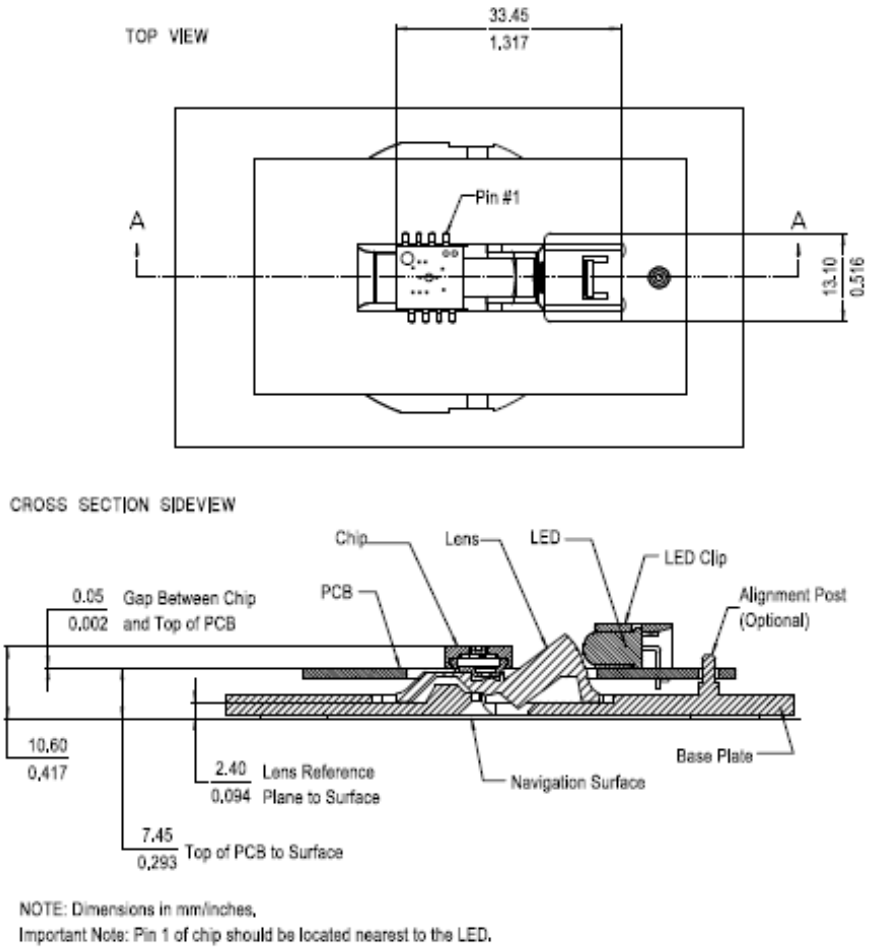


Figure 7. 2D Assembly

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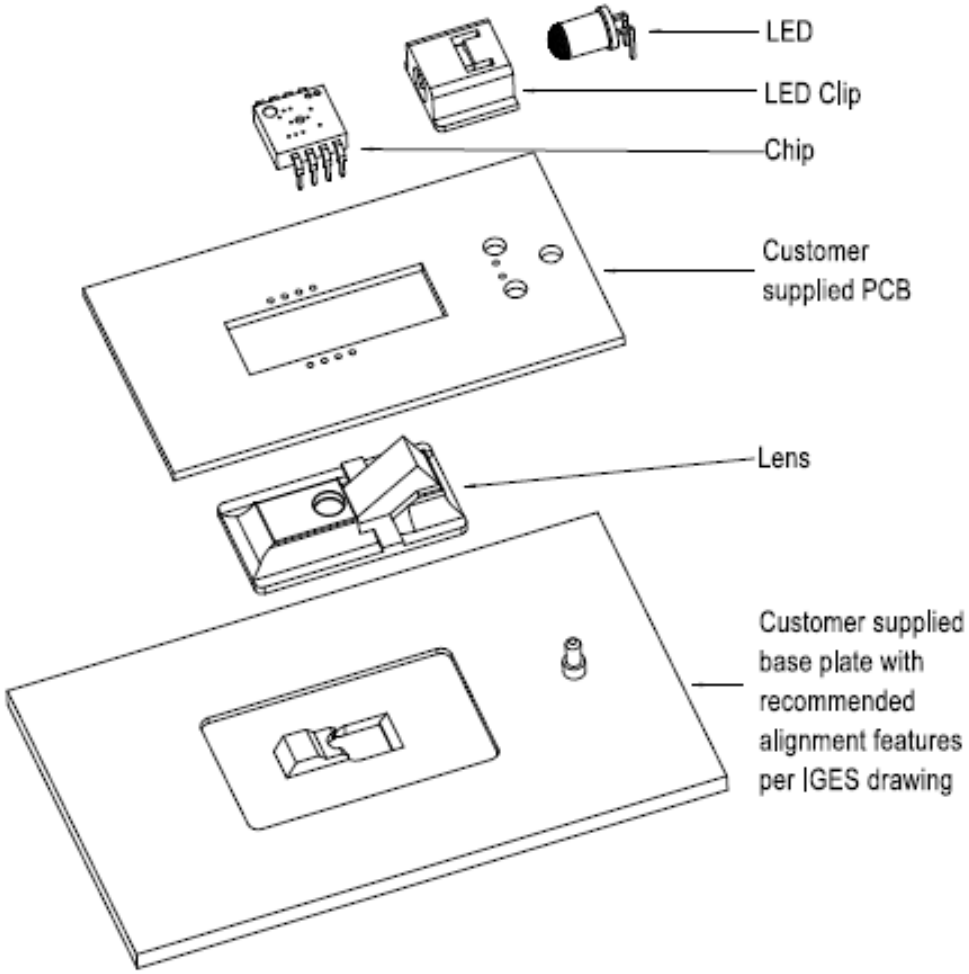
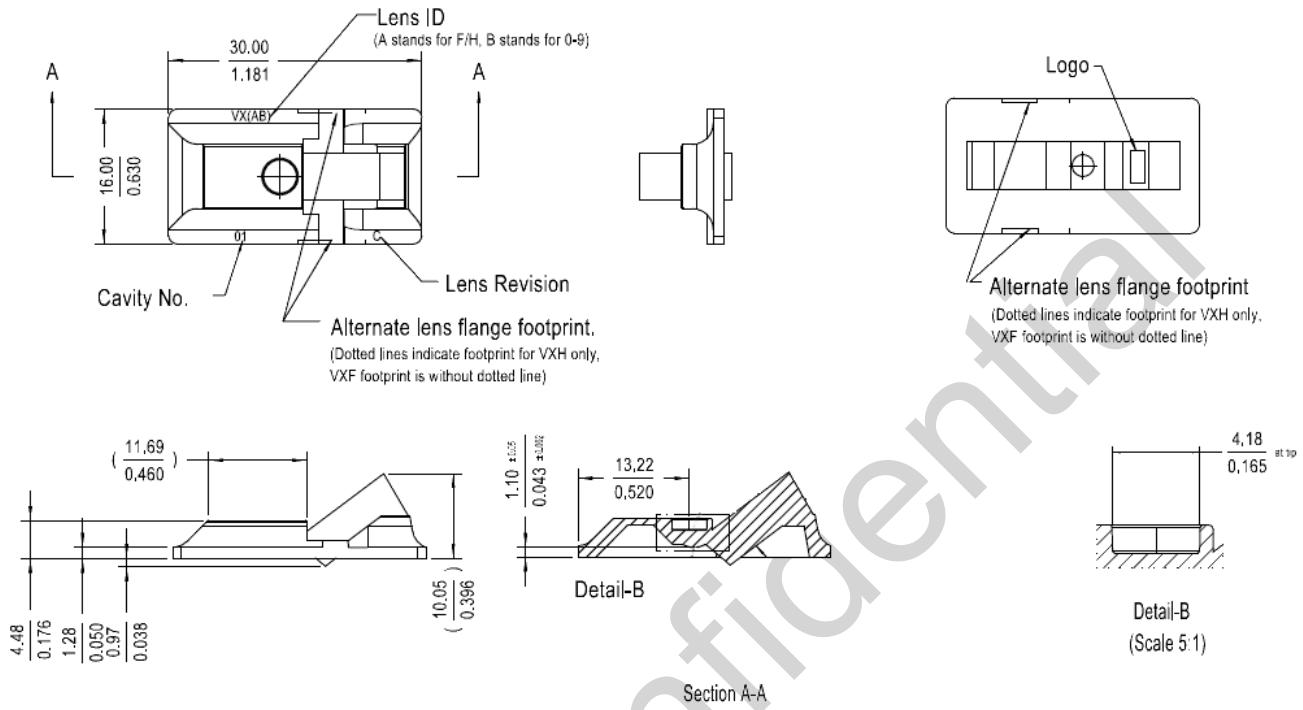


Figure 8. 2D Mouse Assembly

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Notes:
1. Dimensions in milimeter / inches
2. Dimensional tolerance: ± 0.1mm
3. Maximum flash: +0.20mm
4. () reference dimension
5. Optical details of lens not shown.
6. Dotted line indicate alternate lens flange footprint
7. Document number: LM31-LNG-G8_002

Figure 9. LM31-LNG Lens Outline Drawing

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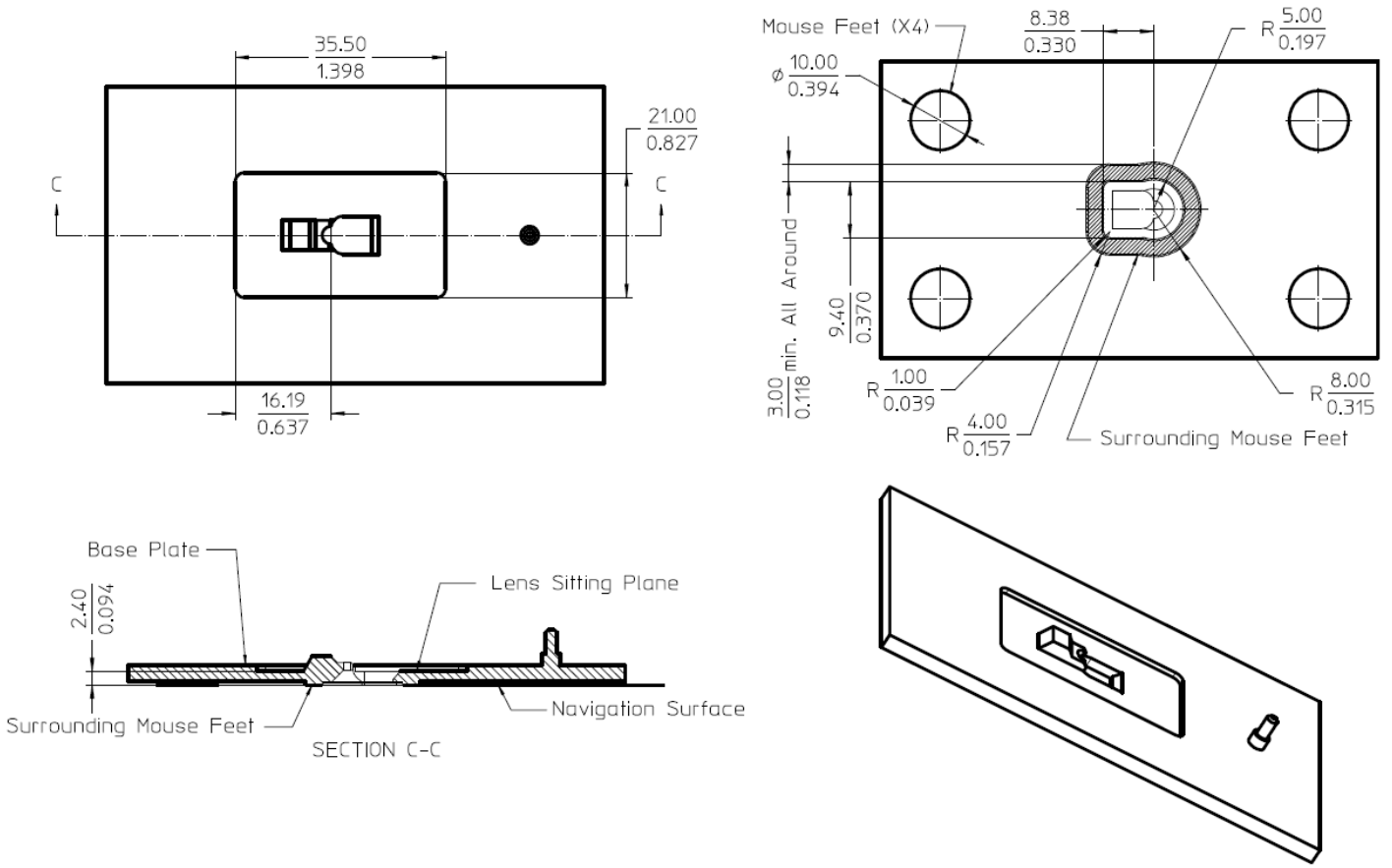


Figure 10. Recommended Base Plate Design with LM31-LNG Lens

Note: Mouse feet should be placed close to the lens opening (hole) to stabilize soft surfaces to be within the working DOF of the chip

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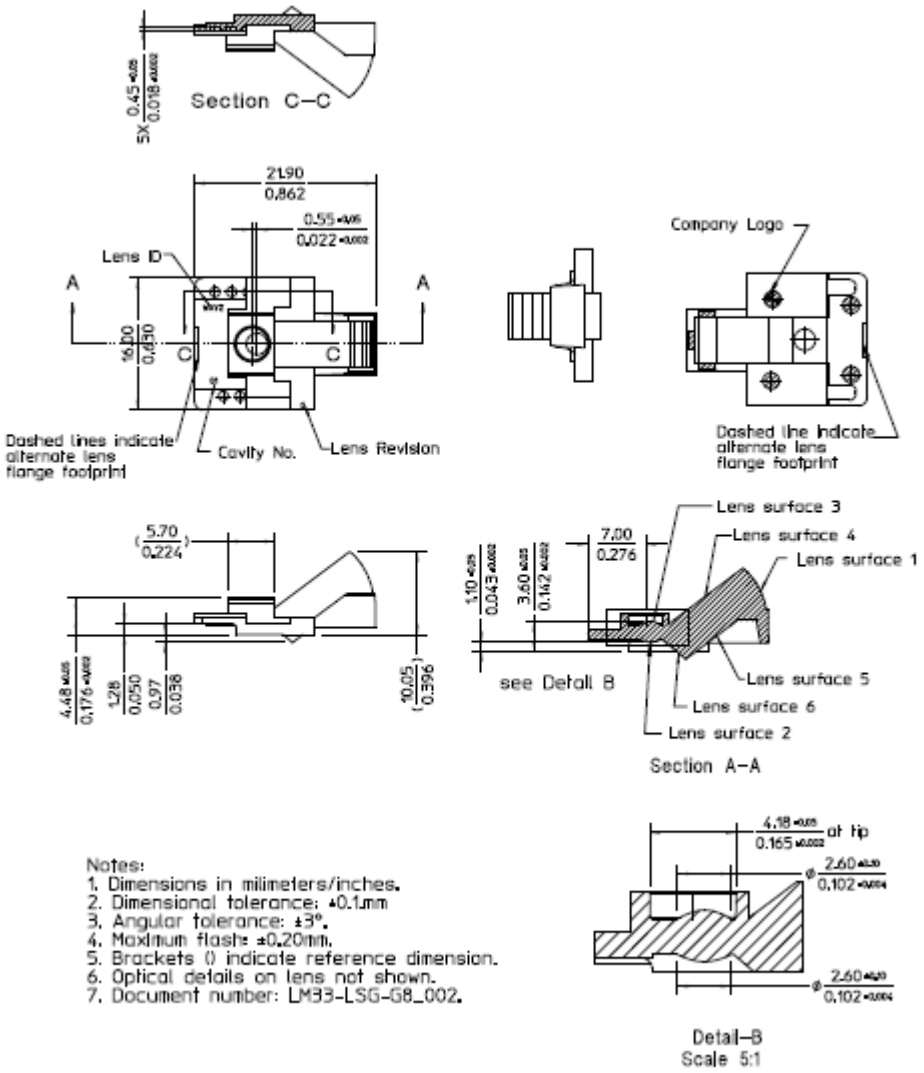


Figure 11. LM33-LSG Lens Outline Drawing

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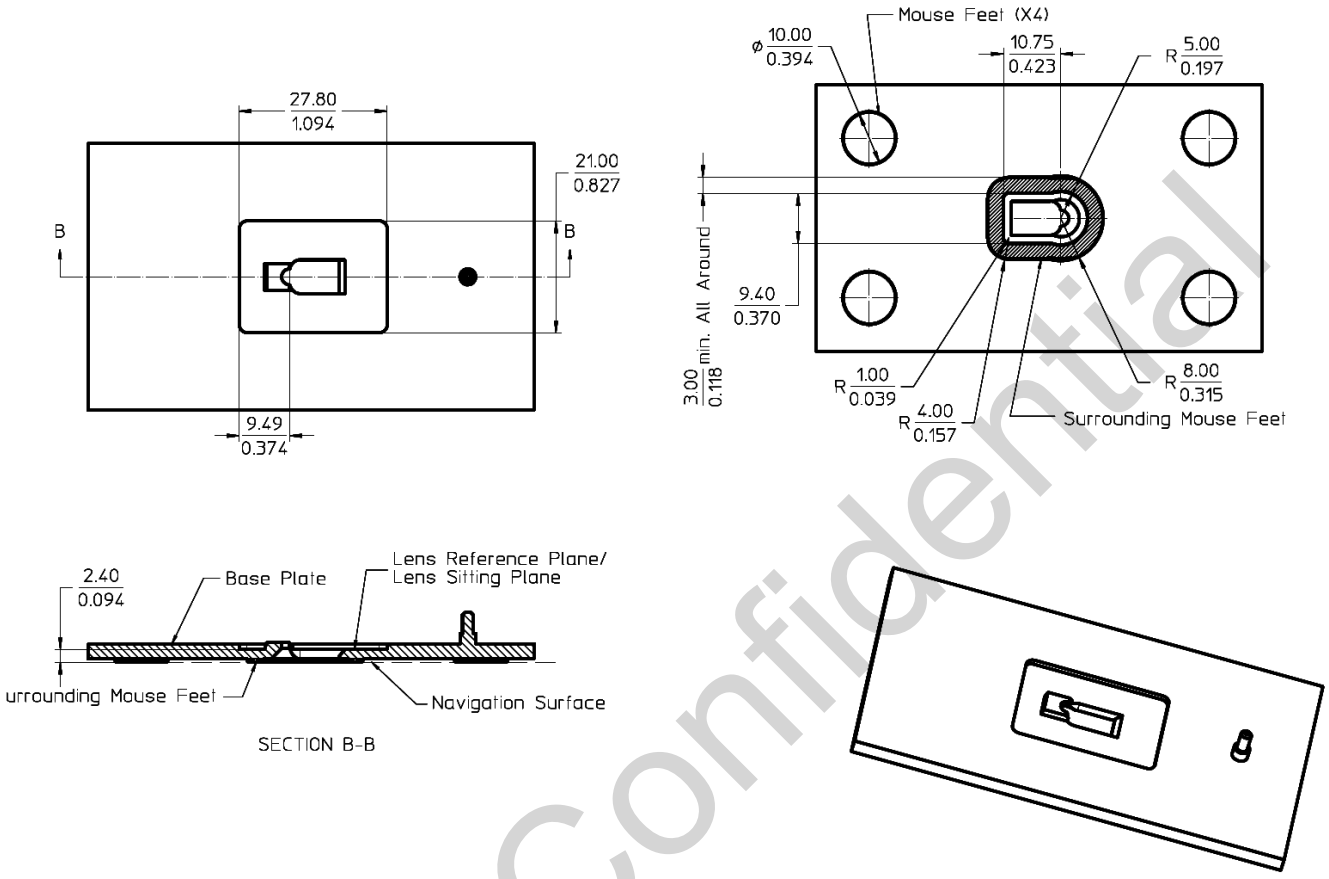


Figure 12. Recommended Base Plate Design with LM33-LSG Lens

Note: Mouse feet should be placed close to the lens opening (hole) to stabilize soft surfaces to be within the working DOF of the chip

3.3 Package Marking

Refer to Figure 2. Pin Configuration

Table 8. Code Identification

Code	Marking	Description
Product Number	PAW3327DB-TWQU	Part number label
Lot Code	YYWWXXXXX	YYWW=Date code XXXXX= PixArt Reserved

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4.0 System Level Description

4.1 Reference Application Schematic Diagram

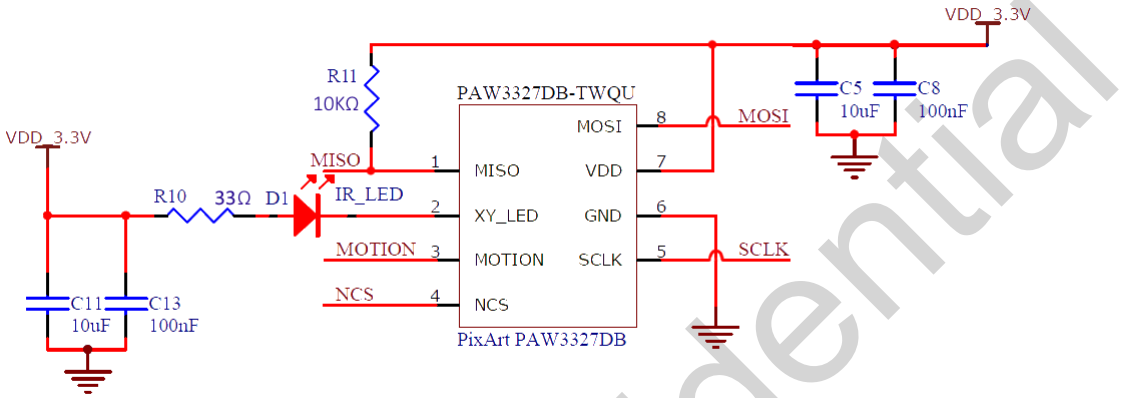


Figure 13. PAW3327DB-TWQU Reference Mouse Schematic

PixArt recommends using the HSDL-4261IR LED with the PAW3327DB-TWQU chip.

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4.2 PCB Assembly Recommendation

1. Insert the integrated chip and all other electrical components into PCB.
2. Wave-solder the entire assembly in a no-wash solder process utilizing solder-fixture. A solder-fixture is required to protect the chip from flux spray and wave solder.
3. Avoid getting any solder flux onto the chip body as there is potential for flux to seep into the chip package. The solder fixture should be designed to expose only the chip leads to flux spray & molten solder while shielding the chip body and optical apertures. The fixture should also set the chip at the correct position and height on the PCB.
4. Place the lens onto the base plate. Care must be taken to avoid contamination on the optical surfaces.
5. Remove the protective kapton tapes from optical apertures of the chip. Care must be taken to prevent contaminants from entering the apertures. Do not place the PCB with the chip facing up during the entire product assembly process. Hold the PCB vertically when removing kapton tape.
6. Remove the protective kapton tapes from optical apertures of the chip. Care must be taken to prevent contaminants from entering the apertures. Do not place the PCB with the chip facing up during the entire mouse assembly process. Hold the PCB vertically when removing kapton tape.
7. Insert PCB assembly over the lens onto the base plate aligning post to retain PCB assembly. The chip package will self-align to the lens via the guide posts. The optical position reference for the PCB is set by the base plate and lens. Note that the PCB motion due to button presses must be minimized to maintain optical alignment.
8. Install mouse top case. There must be a feature in the top case to press down onto the PCB assembly to ensure all components are stacked or interlocked to the correct vertical height

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5.0 Power States & Sequence

5.1 Power-Up Sequence

The PAW3327DB-TWQU performs an internal power up self-reset. The appropriate sequence is as follows:

1. Apply power to VDD.
2. Drive NCS high, and then low to reset the SPI port.
3. Write 0x5A to Power_Up_Reset register.
4. Wait for 50ms.
5. Read from registers 0x02, 0x03, 0x04, 0x05 and 0x06 one time regardless of the motion bit state.
6. Load power up initialization register setting in the following sequence:

   - Write register 0x20 with value 0x40
   - Write register 0x7F with value 0x01
   - Write register 0x1A with value 0x50
   - Write register 0x6F with value 0xFF
   - Write register 0x6C with value 0x3B
   - Write register 0x6D with value 0x2B
   - Write register 0x7A with value 0x4F
   - Write register 0x49 with value 0x40
   - Write register 0x7F with value 0x00
   - Write register 0x4A with value 0x16
   - Write register 0x4D with value 0x53
   - Write register 0x1B with value 0x21
   - Write register 0x62 with value 0x16
   - Write register 0x1A with value 0x11
   - Write register 0x7F with value 0x03
   - Read register 0x1C
   - Read register 0x1A and store in VarA
   - VarB = (((VarA ≫ 1) + 0x23) ^ 0x5A)
   - Write register 0x1B with value in VarB
   - Write register 0x7F with value 0x00
   - Write register 0x1A with value 0x11
   - Write register 0x5D with value 0x81
   - Write register 0x7F with value 0x00

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During power-up there will be a period of time after the power supply is high but before any clocks are available.

The table below shows the state of the various pins during power-up and reset.

Table 9. State of Signal Pins after VDD is Valid

Pin	During Reset	After Reset
NCS	Ignored	Functional and chip is selected when driven low
SCLK	Undefined	Functional when NCS is low
MISO	Ignored	Functional when NCS is low
MOSI	Ignored	Functional when NCS is low
MOTION	Undefined	Functional
XYLED	High	Functional

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5.2 Power Down Sequence

PAW3327DB-TWQU can be shut down by writing to Shutdown register 0x3B with value 0xB6. The SPI port should not be accessed when Shutdown mode is asserted, except the power-up command (writing 0x5A to register 0x3A). Other ICs on the same SPI bus can be accessed, as long as the chip's NCS pin is not asserted.

To de-assert Shutdown mode:

1. Drive NCS high, and then low to reset the SPI port.
2. Write 0x5A to Power_Up_Reset register.
3. Wait for at least 50ms.
4. Read from registers 0x02, 0x03, 0x04, 0x05 and 0x06 one time regardless of the motion bit state.
5. Load power up initialization register setting as per Section 5.0, step 6.

Table 10. Status during Shutdown Mode

Pin	Status during Shutdown Mode
NCS	High *1
SCLK	Ignore if NCS = 1 *2
MISO	Ignore if NCS = 1*3
MOSI	Ignore if NCS = 1
XYLED	High

*1. NCS pin must be held to 1 (high) if SPI bus is shared with other devices. It is recommended to hold to 1 (high) during Shutdown unless powering up the chip. It must be held to 0 (low) if the chip is to be re-powered up from shutdown (writing 0x5A to register 0x3A).
*2. SCLK is ignored if NCS is 1 (high). It is functional if NCS is 0 (low).
*3. MISO should be pulled up during Shutdown in order to meet the low power consumption specification in the data sheet.

Note: There is long wakeup time from shutdown. These features should not be used for power management during normal mouse motion.

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6.0 Serial Peripheral Interface Communication

6.1 Signal Description

The synchronous serial port is used to write and read parameters in the chip. The port is a four wire serial port. The host micro-controller always initiates communication and the chip never initiates communication. SCLK, MOSI & NCS may be driven directly by a micro-controller. The port pins may be shared with other SPI slave devices. When the NCS pin is high, the inputs are ignored and the output is tri-stated.

The lines that comprise the SPI port:

Pin	Description
SCLK	Clock input, generated by the master (microcontroller).
MOSI	Input data (Master Out/Slave In)
MISO	Output data (Master In/Slave Out)
NCS	Chip select input (active low). NCS needs to be low to activate the serial port; otherwise, MISO will be high Z, and MOSI & SCLK will be ignored. NCS can also be used to reset the serial port in case of an error.

6.2 Motion Bit Timing

The motion bit is an output that signals the micro-controller when motion has occurred. The motion bit turns to “1” whenever the motion is detected; in other words, whenever there is non-zero data in the Delta_X_L, Delta_X_H, Delta_Y_L or Delta_Y_H registers. Clearing the motion bit by reading MOTION registers will set the Motion bit back to “0”.

6.3 Chip Select Operation

The serial port is activated after NCS goes low. If NCS is raised during a transaction, the entire transaction is aborted and the serial port will reset. After a transaction is aborted, the normal address-to-data or transaction-to-transaction delay is still required before beginning the next transaction. In order to improve communication reliability, all serial transactions should be framed by NCS. In other words, the port should not remain enabled during periods of non-use because ESD and EFT/B events could be interpreted as serial communication and set the chip into unknown state. In addition, NCS must be raised after each burst-mode transaction is complete to terminate burst-mode. The port is not available for further use until burst-mode is terminated.

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6.4 Write Operation

Write operation is defined as data going from the micro-controller to the chip. It is always initiated by the micro-controller and consists of two bytes. The first byte contains the address (seven bits) and has a "1" as its MSB to indicate write sequence. The second byte contains the data. The chip reads MOSI on the rising edges of SCLK.

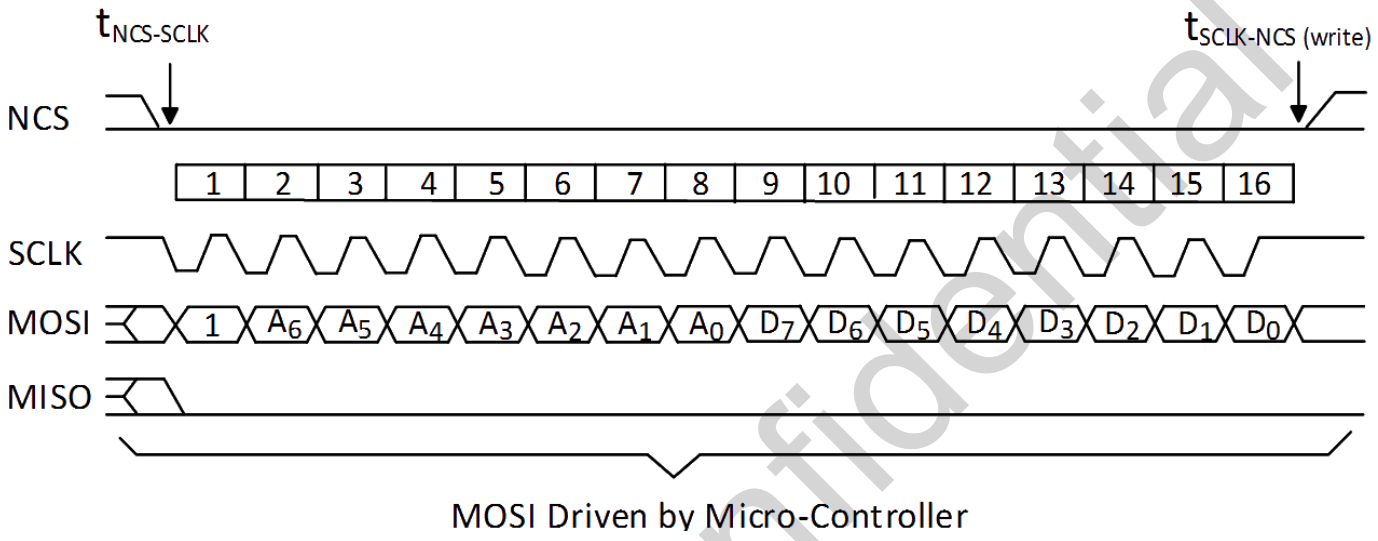


Figure 14. Write Operation

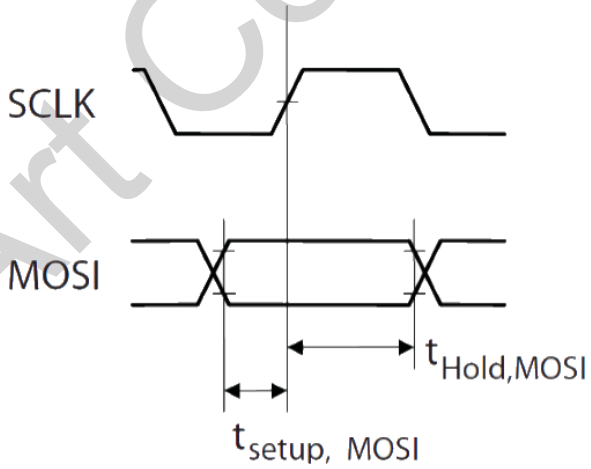


Figure 15. MOSI Setup & Hold Time during write operation

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6.5 Read Operation

A read operation is defined as data going from the chip to the microcontroller. It is always initiated by the microcontroller and consists of two bytes. The first byte contains the address, is sent by the microcontroller over MOSI and has a "0" as its MSB to indicate data direction. The second byte contains the data and is driven by the chip over MISO. The chip outputs MISO bits on falling edges of SCLK and samples MOSI bits on every rising edge of SCLK.

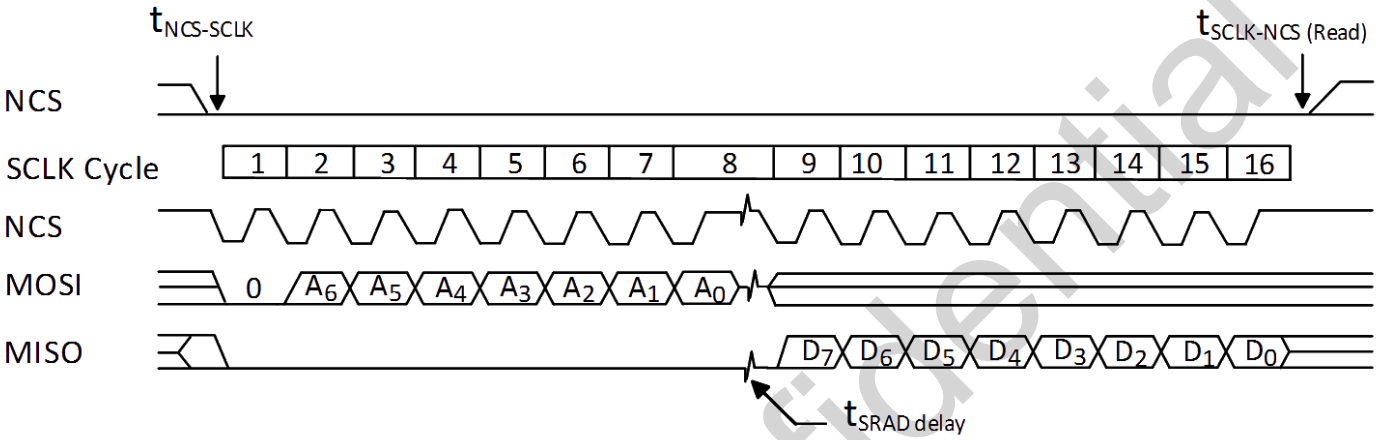


Figure 16. Read Operation

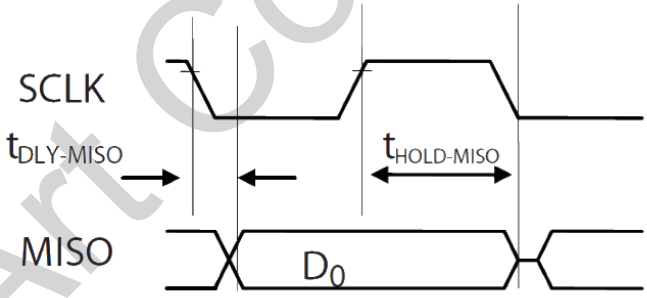


Figure 17. MISO Delay & Hold Time during read operation

Note: The minimum high state of SCLK is also the minimum MISO data hold time of the chip. The falling edge of SCLK is the start of the next read or write command, the chip will hold the state of data on MISO until the falling edge of SCLK.

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6.6 Required Timing between Read and Write Commands

Required timing between Read and Write Commands (t_{SXX}). The NCS should be pulled low for below condition, please refer to the section 6.4 and 6.5 for full timing diagram.

There are minimum timing requirements between read and write commands on the serial port.

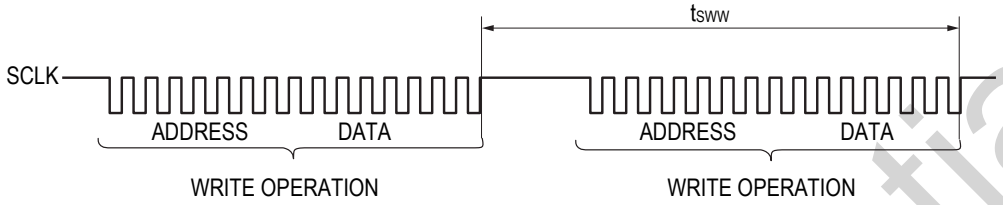


Figure 18. Timing between two Write commands

If the rising edge of the SCLK for the last data bit of the second write command occurs before the required delay (t_{SWW}), then the first write command may not complete correctly.

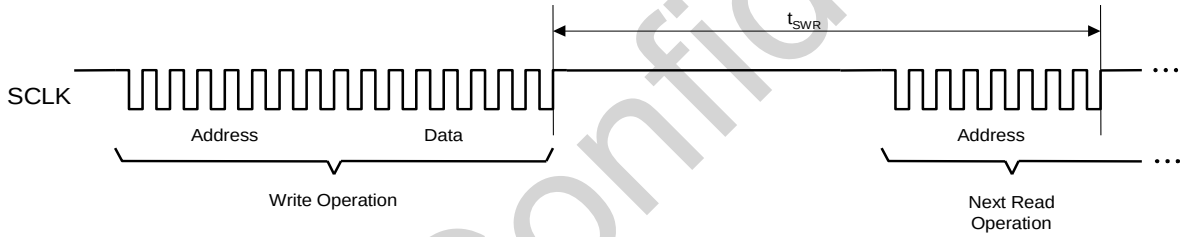


Figure 19. Timing between Write & Read commands

If the rising edge of SCLK for the last address bit of the read command occurs before the required delay (t_{SWR}), the write command may not complete correctly.

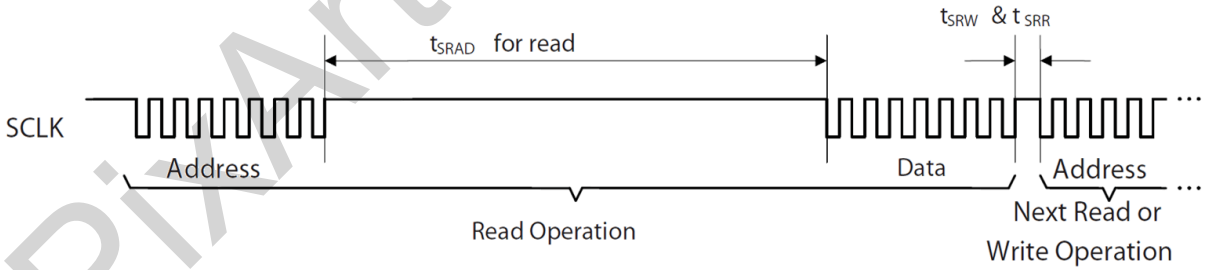


Figure 20. Timing between Read & Subsequent Write or Read commands

If the rising edge of SCLK for the last address bit of the read command occurs before the t_{SWR} required delay, the write command may not be completed correctly. During a read operation SCLK should be delayed at least t_{SRAD} after the last address data bit to ensure that the chip has sufficient time to prepare the requested data.

The falling edge of SCLK for the first address bit of either the read or write command must be at least t_{SRR} or t_{SRW} after the last SCLK rising edge of the last data bit of the previous read operation. In addition, during a read operation SCLK should be delayed after the last address data bit to ensure that the chip has sufficient time to prepare the requested data.

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6.7 Burst Mode Operation

Burst mode is a special serial port operation mode which may be used to reduce the serial transaction time for predefined registers. The speed improvement is achieved by continuous data clocking to or from multiple registers without the need to specify the register address and by not requiring the normal delay period between data bytes.

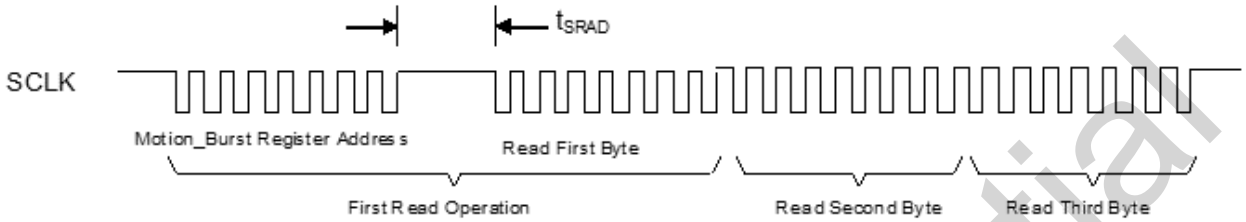


Figure 21. Motion Burst Timing

Motion Burst Read

Reading the Burst_Motion_Read register activates this mode. The chip will respond with the following motion burst report in order. Motion burst report:

BYTE[00] = Motion
BYTE[01] = Observation
BYTE[02] = Delta_X_L
BYTE[03] = Delta_X_H
BYTE[04] = Delta_Y_L
BYTE[05] = Delta_Y_H
BYTE[06] = SQUAL
BYTE[07] = Raw_Data_Sum
BYTE[08] = Maximum_Raw_Data
BYTE[09] = Minimum_Raw_Data
BYTE[10] = Shutter_Upper
BYTE[11] = Shutter_Lower

After sending the register address, the microcontroller must wait for t_{SRAD}, and then begin reading data. All data bits can be read with no delay between bytes by driving SCLK at the normal rate. The data are latched into the output buffer after the last address bit is received. After the burst transmission is complete, the microcontroller must raise the NCS line for at least t_{BEXIT} to terminate burst mode. The serial port is not available for use until it is reset with NCS, even for a second burst transmission.

Procedure to start motion burst as below.

1. Pull NCS line to low.
2. Read Motion_Burst register (address 0x16).
3. Wait for t_{SRAD}.
4. Start reading SPI Data continuously up to 12 bytes.
   Motion burst may be terminated by pulling NCS high for at least t_{BEXIT}.
5. To read new motion burst data, start again from step 1.

Burst mode must be terminated by the micro-controller by raising the NCS line for at least t_{BEXIT}. The serial port is not available for use until it is reset with NCS signal, even for a second Burst transmission.

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7.0 Raw Data Grab

Raw data grab is the way to download a full array of raw data values from a single frame. Once raw data grab is enabled, the next complete frame image will be stored to memory. In order to stream out the Raw Data values, register read operation is used.

Procedure to start frame capture burst mode is listed as below:

1. Write register 0x59 with value 0x00.
2. Write register 0x5C with value 0x61.
3. Wait for 25ms.
4. Write register 0x6B with value 0x80.
5. Write register 0x32 with value 0xFF.
6. Read register 0x19 until both PG_VALID and PG_FIRST bits are set to "1".
7. Read register 0x32 for 625 times with at least 1 frame period (recommended 500us) delay between reads. The full set of 7-bit ADC data (RAWDATA bits [6:0]) is used to form one complete picture array. Ensure that valid bit [7] is set to "1" for every raw data grab readout. Data with valid bit set to "0" indicate error and raw data grab process would need to restart from step (5).
8. Write register 0x6B with value 0x00.
9. Write register 0x5C with value 0x41.
10. Write register 0x59 with value 0x0F.

Last Raw Data

24	49	74	99	124	149	174	199	224	249	274	299	324	349	374	399	424	449	474	499	524	549	574	599	624
23	48	73	98	123	148	173	198	223	248	273	298	323	348	373	398	423	448	473	498	523	548	573	598	623
22	47	72	97	122	147	172	197	222	247	272	297	322	347	372	397	422	447	472	497	522	547	572	597	622
-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
2	27	52	77	102	127	152	177	202	227	252	277	302	327	352	377	402	427	452	477	502	527	552	577	602
1	26	51	76	101	126	151	176	201	226	251	276	301	326	351	376	401	426	451	476	501	526	551	576	601
0	25	50	75	100	125	150	175	200	225	250	275	300	325	350	375	400	425	450	475	500	525	550	575	600

First Raw Data

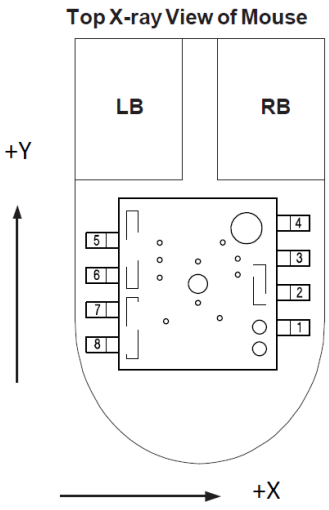


Figure 22. Raw Data Address Map for 25x25 (Chip looking on the navigation surf

***Note**: The X/Y reporting direction shown above follows all the power up initialization sequence in the Power Up section.*

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8.0 Registers

8.1 Registers List

The chip registers are accessible via the serial port. The registers are used to read motion data and status as well as to set the device configuration.

Table 11. Register List

Address	Register Name	Access	Reset Value
0x00	PRODUCT_ID	R	0x4C
0x02	MOTION	RW	0x00
0x03	DELTA_X_L	R	0x00
0x04	DELTA_X_H	R	0x00
0x05	DELTA_Y_L	R	0x00
0x06	DELTA_Y_H	R	0x00
0x07	SQUAL	R	0x00
0x08	RAW_DATA_SUM	R	0x00
0x09	MAXIMUM_ RAW_DATA	R	0x00
0x0A	MINIMUM_ RAW_DATA	R	0x00
0x0B	SHUTTER_LOWER	R	0x12
0x0C	SHUTTER_UPPER	R	0x00
0x15	CHIP_OBSERVATION	RW	0x80
0x16	BURST_MOTION_READ	R	0x00
0x19	RAW_DATA _GRAB_STATUS	R	0x00
0x1A	RIPPLE_CONTROL	RW	0x10
0x1B	Resolution	RW	0x21
0x1E	ANGLE_SNAP	RW	0x04
0x20	AXIS_CONTROL	RW	0x00
0x32	RAW_DATA_GRAB	RW	0x00
0x3A	POWER_UP_RESET	W	N/A
0x3B	SHUTDOWN	W	N/A
0x3F	INV_PRODUCT_ID	R	0xB3

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8.2 Register Descriptions

Register Name	Product_ID							
Address	0x00							
Access	Read			Reset Value		0x4C		
Bit	7	6	5	4	3	2	1	0
Field	PID_{7-0}							
Description	This register contains a unique identification assigned to the PAW3327DB-TWQU. The value in this register does not change; it can be used to verify that the serial communications link is functional.							

Register Name	Motion							
Address	0x02							
Access	Read/Write			Reset Value		0x00		
Bit	7	6	5	4	3	2	1	0
Field	MOT	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved

Description

This register allows the user to determine if motion has occurred since the last time it was read.

The procedure to read the motion registers (Delta_X_L, Delta_X_H, Delta_Y_L and Delta_Y_H) is as follows:

1. Read the Motion register. This will freeze the Delta_X_L, Delta_X_H, Delta_Y_L and Delta_Y_H register values.

2. If the MOT bit is set, Delta_X_L, Delta_X_H, Delta_Y_L and Delta_Y_H registers should be read in the given sequence to get the accumulated motion. Note: if Delta_X_L, Delta_X_H, Delta_Y_L and Delta_Y_H registers are not read before the motion register is read for the second time, the data in Delta_X_L, Delta_X_H, Delta_Y_L and Delta_Y_H will be lost.

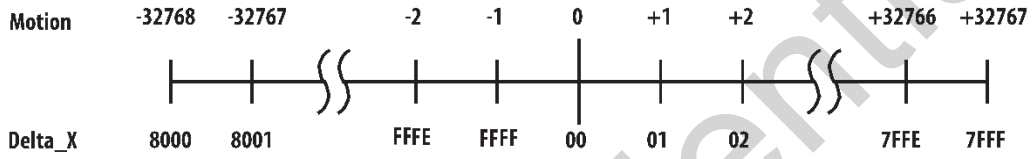
3. To read a new set of motion data (Delta_X_L, Delta_X_H, Delta_Y_L and Delta_Y_H), repeat from Step 1.

Field Name	Description
MOT	Motion since last report **0 = No motion** 1 = Motion occurred, data ready for reading in Delta_X_L, Delta_X_H, Delta_Y_L and Delta_Y_H registers

Write any value to this register will clear all motion data.

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Register Name	DELTA_X_L							
Address	0x03							
Access	Read			Reset Value		0x00		
Bit	7	6	5	4	3	2	1	0
Field	X_7	X_6	X_5	X_4	X_3	X_2	X_1	X_0
Description	16 bits 2's complement number. Lower 8 bits of Delta_X. X movement is counts since last report. Absolute value is determined by resolution.  							

Register Name	DELTA_X_H							
Address	0x04							
Access	Read			Reset Value		0x00		
Bit	7	6	5	4	3	2	1	0
Field	X_{15}	X_{14}	X_{13}	X_{12}	X_{111}	X_{10}	X_9	X_8
Description	16 bits 2's complement number. Upper 8 bits of Delta_X. Delta_X_H must be read after Delta_X_L to have the full motion data. ***Note:*** *It is recommended that register 0x02, 0x03, 0x04, 0x05 and 0x06 to be read sequentially.*							

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Register Name	DELTA_Y_L							
Address	0x05							
Access	Read			Reset Value		0x00		
Bit	7	6	5	4	3	2	1	0
Field	Y_7	Y_6	Y_5	Y_4	Y_3	Y_2	Y_1	Y_0
Description	16 bits 2's complement number. Lower 8 bits of Delta_Y. Y movement is counts since last report. Absolute value is determined by resolution.  							

Register Name	DELTA_Y_H							
Address	0x06							
Access	Read			Reset Value		0x00		
Bit	7	6	5	4	3	2	1	0
Field	Y_{15}	Y_{14}	Y_{13}	Y_{12}	Y_{11}	Y_{10}	Y_9	Y_8
Description	16 bits 2's complement number. Upper 8 bits of Delta_Y. Delta_Y_H must be read after Delta_Y_L to have the full motion data. ***Note:*** *It is recommended that register 0x02, 0x03, 0x04, 0x05 and 0x06 to be read sequentially.*							

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Register Name	SQUAL							
Address	0x07							
Access	Read			Reset Value		0x00		
Bit	7	6	5	4	3	2	1	0
Field	SQ_7	SQ_6	SQ_5	SQ_4	SQ_3	SQ_2	SQ_1	SQ_0

Description

The SQUAL (Surface quality) register is a measure of the number of valid features visible by the chip in the current frame. Use the following formula to find the total number of valid features.

Number of Features = SQUAL Register Value * 4

The maximum SQUAL register value is 0x9C. Since small changes in the current frame can result in changes in SQUAL, variations in SQUAL when looking at a surface is expected. The graph below shows 500 sequentially acquired SQUAL values, while a chip was moved slowly over white paper.

SQUAL values are only valid when chip is in run mode. Disable Rest mode before measuring SQUAL.

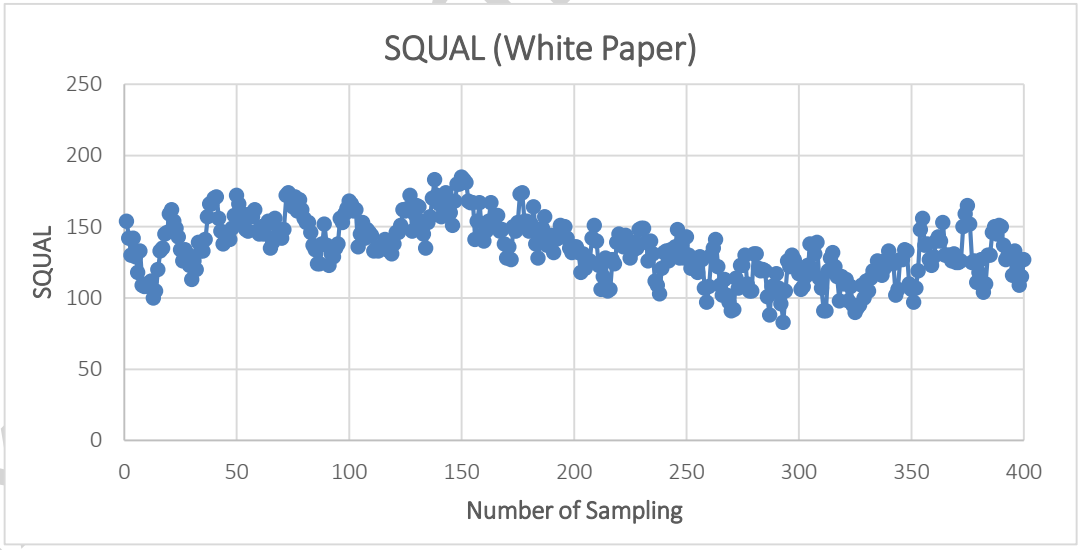


Figure 23. SQUAL values on White Paper

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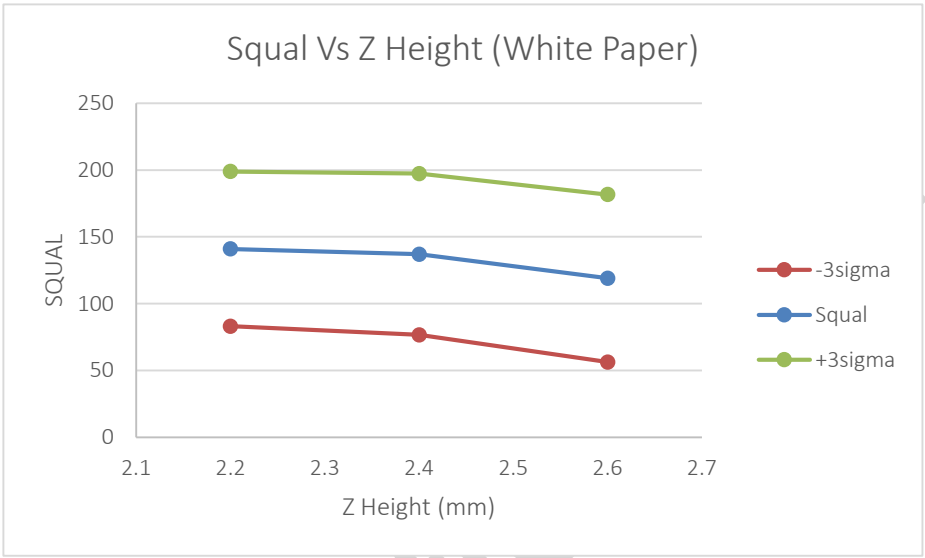


Figure 24. Mean SQUAL vs. Z Height (White Paper)

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Register Name	RAW_DATA_SUM							
Address	0x08							
Access	Read			Reset Value		0x00		
Bit	7	6	5	4	3	2	1	0
Field	AP_7	AP_6	AP_5	AP_4	AP_3	AP_2	AP_1	AP_0
Description	This register is used to find the chip average raw data value. It reports the upper byte of a 17-bit counter which sums all 625 raw data in the current frame. The maximum register value is 0x9B. The minimum register value is 0. The raw data sum value can change every frame.							

Register Name	MAXIMUM_RAW_DATA							
Address	0x09							
Access	Read			Reset Value		0x00		
Bit	7	6	5	4	3	2	1	0
Field	MP_7	MP_6	MP_5	MP_4	MP_3	MP_2	MP_1	MP_0
Description	Maximum Raw Data value in current frame. Minimum value = 0, maximum value = 127. The maximum raw data value can change every frame.							

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Register Name	MINIMUM_RAW_DATA							
Address	0x0A							
Access	Read			Reset Value		0x00		
Bit	7	6	5	4	3	2	1	0
Field	$MinP_7$	$MinP_6$	$MinP_5$	$MinP_4$	$MinP_3$	$MinP_2$	$MinP_1$	$MinP_0$
Description	Minimum Raw data value in current frame. Minimum value = 0, maximum value = 127. The minimum raw data value can change every frame.							

Register Name	SHUTTER_LOWER							
Address	0x0B							
Access	Read			Reset Value		0x12		
Bit	7	6	5	4	3	2	1	0
Field	S_7	S_6	S_5	S_4	S_3	S_2	S_1	S_0
Description	Lower byte of the 12-bit Shutter register.							

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Register Name	SHUTTER_UPPER							
Address	0x0C							
Access	Read			Reset Value		0x00		
Bit	7	6	5	4	3	2	1	0
Field	Reserved	Reserved	Reserved	S_{12}	S_{11}	S_{10}	S_9	S_8
Description	Upper 4-bit of the 12-bit Shutter register. Units are clock cycles of the internal oscillator. Read Shutter_Upper first, then Shutter_Lower. They should be read consecutively. The shutter is adjusted to keep the average raw data values within normal operating ranges. The shutter value is checked and automatically adjusted to a new value if needed on every frame when operating in default mode.							

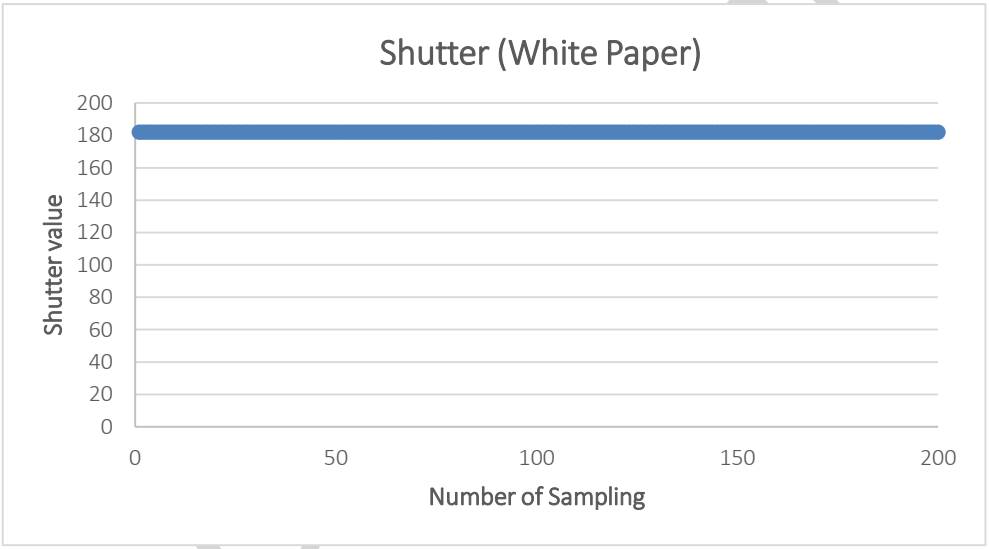


Figure 25. Shutter Values on White Paper

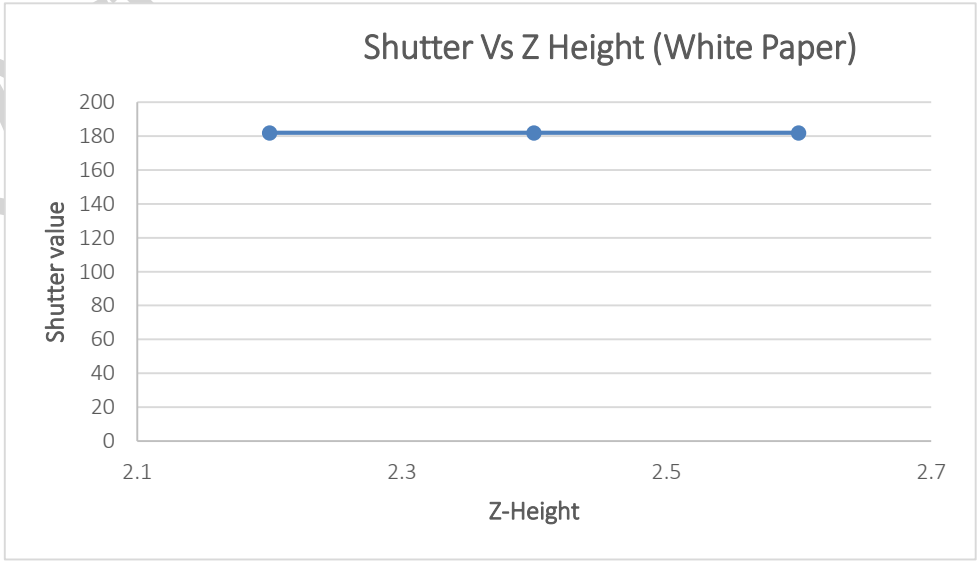


Figure 26. Mean Shutter vs. Z Height (White Paper)

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Register Name	CHIP_OBSERVATION							
Address	0x15							
Access	Read/Write			Reset Value		0x80		
Bit	7	6	5	4	3	2	1	0
Field	Reserved	Reserved	CO_5	CO_4	CO_3	CO_2	CO_1	CO_0
Description	The user must clear the register by writing 0x00, wait for a minimum T_{dly_obs} ms & read the register. The active process CO_{5-0} will have set the corresponding bit. The register may be used as part of recovery scheme to detect a problem caused by EFT/B or ESD event. T_{dly_obs} is defined as the longest frame processing time. Clock frequency tolerance value need to be taken into account. Therefore, T_{dly_obs} = 1.5ms.							

Register Name	BURST_MOTION_READ							
Address	0x16							
Access	Read			Reset Value		0x00		
Bit	7	6	5	4	3	2	1	0
Field	MB_7	MB_6	MB_5	MB_4	MB_3	MB_2	MB_1	MB_0
Description	The Burst_Motion_Read register is used for high-speed access to the Motion, Observation, Delta_X_L, Delta_X_H, Delta_Y_L, Delta_Y_H, SQUAL, Raw Data_Sum, Maximum_Raw_Data, Minimum_Raw_Data, Shutter_Upper and Shutter_Lower registers. See Burst Mode-Motion Read section 7.7 for use details.							

Register Name	RAW_DATA_GRAB_STATUS							
Address	0x19							
Access	Read			Reset Value		0x00		
Bit	7	6	5	4	3	2	1	0
Field	Reserved	PG_VALID	PG_FIRST	Reserved	Reserved	Reserved	Reserved	Reserved
Description	This register provides additional information for user to monitor the chip navigation status.							

Field Name	Description
PG_VALID	1 = Raw Data Grab valid
PG_FIRST	1 = Raw Data Grab first

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Register Name	RIPPLE_CONTROL							
Address	0x1A							
Access	Read/Write			Reset Value		0x10		
Bit	7	6	5	4	3	2	1	0
Field	Reserved	Reserved	Reserved	1	Reserved	Reserved	Reserved	EN

Description

Set bit 0 to enable ripple control.

Field Name	Description
EN	0: Ripple Control Disable 1: Ripple Control Enable

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Register Name	RESOLUTION							
Address	0x1B							
Access	Read/Write			Reset Value		0x21		
Bit	7	6	5	4	3	2	1	0
Field	RES_7	RES_6	RES_5	RES_4	RES_3	RES_2	RES_1	RES_0

Description

This register provides additional information for user to monitor the chip navigation status.

This register set the resolution XY of the chip. The approximate resolution value for each register setting can be defined using the chart shown below.

Field Name	Description
$RES_{7:0}$	Set the chip resolution with 100cpi step of approximately, starting from minimum of 200cpi.

Hex	Target	Hex	Target	Hex	Target
0x04	200	0x3E	2700	0x79	5200
0x06	300	0x40	2800	0x7B	5300
0x09	400	0x42	2900	0x7E	5400
0x0B	500	0x45	3000	0x80	5500
0x0E	600	0x48	3100	0x82	5600
0x10	700	0x4A	3200	0x85	5700
0x12	800	0x4D	3300	0x87	5800
0x15	900	0x4E	3400	0x89	5900
0x17	1000	0x51	3500	0x8C	6000
0x19	1100	0x54	3600	0x8E	6100
0x1B	1200	0x55	3700	0x91	6200
0x1D	1300	0x58	3800		
0x20	1400	0x5B	3900		
0x22	**1500**	0x5D	4000		
0x24	1600	0x5F	4100		
0x27	1700	0x62	4200		
0x29	1800	0x64	4300		
0x2B	1900	0x66	4400		
0x2E	2000	0x69	4500		
0x30	2100	0x6B	4600		
0x32	2200	0x6D	4700		
0x34	2300	0x70	4800		
0x37	2400	0x72	4900		
0x39	2500	0x74	5000		
0x3B	2600	0x77	5100		

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Register Name	ANGLE_SNAP							
Address	0x1E							
Access	Read/Write			Reset Value		0x04		
Bit	7	6	5	4	3	2	1	0
Field	EN	0	0	0	0	1	0	0

Description

Write to this register to enable angle snap feature.

Field Name	Description
EN	0: Angle snap disable 1: Angle snap enable

Register Name	AXIS_CONTROL							
Address	0x20							
Access	Read/Write			Reset Value		0x00		
Bit	7	6	5	4	3	2	1	0
Field	Swap_XY	INV_Y	INV_X	Reserved	Reserved	Reserved	Reserved	Reserved

Description

The register set the axis direction of the chip reporting.

Field Name	Description
Swap_XY	1: Swap XY directions
INV_Y	1: Invert Y direction
INV_X	1: Invert X direction

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Register Name	RAW_DATA_GRAB							
Address	0x32							
Access	Read/Write			Reset Value		0x00		
Bit	7	6	5	4	3	2	1	0
Field	VALID	$RAWDATA_6$	$RAWDATA_5$	$RAWDATA_4$	$RAWDATA_3$	$RAWDATA_2$	$RAWDATA_1$	$RAWDATA_0$
Description	This register contains the raw data levels when the Raw Data Grab process is enabled. Refer to section 7.0 for raw data grab usage flow.							

Register Name	POWER_UP_RESET							
Address	0x3A							
Access	Write			Reset Value		NA		
Bit	7	6	5	4	3	2	1	0
Field	$PRST_7$	$PRST_6$	$PRST_5$	$PRST_4$	$PRST_3$	$PRST_2$	$PRST_1$	$PRST_0$
Description	Write 0x5A to this register to reset the chip and all settings will revert to default values. Reset is required after recovering from Shutdown mode.							

Register Name	SHUTDOWN							
Address	0x3B							
Access	Write			Reset Value		NA		
Bit	7	6	5	4	3	2	1	0
Field	SD_7	SD_6	SD_5	SD_4	SD_3	SD_2	SD_1	SD_0
Description	Write 0xB6 to set the chip to Shutdown mode. Refer to the Shutdown section for more details on recovery procedure.							

Register Name	INV_PRODUCT_ID							
Address	0x3F							
Access	Read			Reset Value		0xB3		
Bit	7	6	5	4	3	2	1	0
Field	$IPID_7$	$IPID_6$	$IPID_5$	$IPID_4$	$IPID_3$	$IPID_2$	$IPID_1$	$IPID_0$
Description	This register value is the inverse of the Product_ID register value. It is use to test the SPI port hardware.							

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9.0 Document Revision History

Revision Number	Date	Description
1.00	24 Oct 2017	Official Release Version
1.01	11 Dec 2017	Pg22 - update power up sequence, remove “Read register 0x1C”
1.02	20 Dec 2017	Pg22 – add “Write register 0x7F with value 0x00” Pg42 - corrected typo from "0x28" to "0x2B" and from "3F00" to "3300"
1.10	3 Oct 2018	Pg22 – update power up sequence Pg42 – update resolution table

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